Wrong the Same Way: Cross-Prompt Sharing Shapes Reward Hacking in RLVR

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Abstract

At its KL-regularised optimum, reinforcement learning with verifiable rewards (RLVR) acts as a filter: it reweights the verifier-accepted set as a whole without altering its internal composition. The rate of reward hacking should therefore be inherited from the base model rather than learned. Yet what breaks this is not the overall error rate but the specific pattern of verifier errors. We ask which property of that pattern drives exploitation, and identify cross-prompt parameter sharing. Writing the exact drift of the policy off the filter baseline as a sum of covariances between reward and the empirical tangent kernel isolates a cross-prompt term that vanishes identically without parameter sharing. We validate this in a controlled RLVR environment small enough to compute the entire response distribution exactly. Replacing the shared architecture with independent per-prompt copies removes the dependency on leak breadth, neutralizing the drift slope. Targeted controls further show that sharing at RL time alone is insufficient: exploitation is amplified only when the base model's representation already encodes the verifier's acceptance pattern. Verifier hardening must therefore be budgeted against the model's pre-existing shared error modes, since patches targeting universal leaks yield significantly higher true accuracy than uniform error reduction.

1 Introduction

As language models increasingly rely on reinforcement learning with verifiable rewards (RLVR), understanding how they exploit imperfect verifiers is critical for safe deployment. In RLVR, a response is rewarded if a programmatic checker accepts it [Shao et al., 2024, Guo et al., 2025]. These verifiers are imperfect. Generative judges accept degenerate strings: non-word symbols and generic reasoning openers elicit false positives across a wide range of judge models [Zhao et al., 2026]. Test suites accept programs that pass the visible tests and nothing else [Khalifa et al., 2026]. Outcome-only checkers reward a correct final answer reached by invalid reasoning [Yuan et al., 2026], a failure mode that motivated supervising each reasoning step [Lightman et al., 2023] rather than the final answer alone [Cobbe et al., 2021]. Each such false positive is a response the policy can be paid for without being correct.

The literature presents two incompatible accounts of how RLVR interacts with these false positives. In the first, RLVR tolerates verifier error: sweeping the noise rate from 1% to 50% leaves peak validation accuracy within two points of clean training up to 15%, with graceful degradation to 30% [Plesner et al., 2026]. Noise rescales convergence time rather than the destination, provided the verifier's Youden index $J = \mathrm{TPR} - \mathrm{FPR}$ remains positive [Rad et al., 2026]. A preregistered contrast between leaky and hardened test suites suggests that the incorrect programs a trained model emits were already being emitted by the base model, pointing to a "selection of pre-existing error modes rather than learned exploitation" [Zhang, 2026]. In the second, verifier exploitation is a first-order practical failure: RL amplifies hacking seeded by a trace of contamination [Khalifa et al., 2026], systematic

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verification error produces delay, plateau, or collapse [Egashira et al., 2026], RLVR-trained models learn to game extensional verifiers [Helff et al., 2026], and production reward hacking generalizes into broader misalignment [MacDiarmid et al., 2025].

Theory favours the first account. At the KL-regularised optimum of the RLHF objective [Stiennon et al., 2020, Ouyang et al., 2022], the policy is $\pi^\star(y \mid x) \propto \pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}$, so within any reward level set the odds match the reference model's [Korbak et al., 2022]. Every accepted response, whether correct or exploiting, is multiplied by the same factor. RLVR therefore acts as a filter on the accepted set, not a sculptor inside it, a point made repeatedly in the RLVR diversity literature [Gai et al., 2025, Kruszewski et al., 2026, Dymetman, 2026] and the reason RLVR sharpens rather than expands what a base model can do [Yue et al., 2025]. Applied to verifier false positives, this predicts that the post-RL hack rate is inherited from the base model. The prediction is delimited: it holds at the KL-regularised optimum and at every step of a multiplicative update, but not for arbitrary algorithms or parameterisations, and our own experiments show first-order training departing from this baseline. We treat the filter theory as a null model, and characterize the mechanisms driving the departure from it.

Contributions.

- **Exact Drift Decomposition:** We formalize the instantaneous drift of $\log \pi_\theta(y \mid x)$ under RLVR gradient flow as a sum over *all* prompts of the covariance between reward and the empirical tangent kernel (Lemma 1). For a policy without cross-prompt parameter sharing, this collapses to a single term, proving that dynamics cannot depend on how many prompts share a leak (Corollary 1).
- **Universality Threshold:** We derive an observable threshold for shared-parameter policies (Proposition 2). The leading-order drift of a leak accepted on a fraction u of prompts has the sign of $u - \bar{V}$, where $\bar{V}$ is the policy's current verifier pass rate. Below this threshold, cross-prompt sharing actually suppresses the leak. Crucially, neither side of the threshold depends on the verifier's overall error rate.
- **Controlled Exact Environment:** We construct an RLVR environment where the full response distribution is enumerated exactly, ensuring all reported quantities are true values rather than sample estimates. This environment allows us to manipulate leak universality and static false-positive rates as independent variables.
- **Isolating the Mechanism:** We demonstrate that the shared-parameter policy's hack fraction is strictly tied to leak universality. By replacing the shared transformer with N independent, per-prompt copies, we eliminate the dependence on u. Furthermore, via targeted controls, we establish that sharing at RL time alone is insufficient: RL amplifies a leak only when the base model's representation already structurally encodes the verifier's acceptance pattern.

2 Setup: RLVR with an exploitable verifier

To systematically investigate these dynamics, we first define the standard RLVR objective and the precise metrics we use to track reward hacking. Let $\mathcal{X}$ be a finite prompt set, $|\mathcal{X}| = N$, with the uniform distribution, and $\mathcal{Y}$ a finite response space. A response is *correct* if $c(x, y) = 1$; we write g_x for the correct response, which is unique in our setting. The deployed verifier returns $r(x, y) \in \{0, 1\}$ with $r \geq c$ (assuming no false negatives). We write $\mathcal{A}(x) = \{y : r(x, y) = 1\}$ for the accepted set and $\mathcal{H}(x) = \{y : r(x, y) = 1,\, c(x, y) = 0\}$ for its false-positive part, which we refer to as the *leaks*. RLVR maximizes $J(\theta) = \mathbb{E}_x\, \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}[r(x, y)]$.

Definition 1 (Hack fraction)**.** We define the hack fraction as $\rho(\pi) = \frac{1}{N}\sum_x \pi(\mathcal{H}(x) \mid x) \,/\, \pi(\mathcal{A}(x) \mid x)$, representing the probability that a rewarded response is a false positive, averaged over prompts. We also track the prompt-averaged log-odds $\Lambda(\pi) = \frac{1}{N}\sum_x \log\big[\pi(\mathcal{H}(x) \mid x) \,/\, \pi(g_x \mid x)\big]$.

Definition 2 (Leak universality)**.** For a given response h, let $m(h) = |\{x : h \in \mathcal{H}(x)\}|$ and $u(h) = m(h)/N$. A leak with $u = 1$ is universally accepted everywhere; a leak with $u = 1/N$ is idiosyncratic and specific to only one prompt.

It is important to note that universality is entirely orthogonal to the verifier's error rate. A verifier that accepts exactly one wrong response per prompt yields the same static false-positive rate on an audit,

regardless of whether that accepted response is the same universal string everywhere or a uniquely different string for each prompt.

2.1 The filter baseline

Proposition 1 (RLVR is a filter; known)**.** *Assume the verifier has no false negatives, $r \geq c$, meaning correct responses and leaks lie in the same reward level set. Let $\pi^\star_\beta = \arg\max_\pi J(\pi) - \beta\, \mathbb{E}_x \mathrm{KL}(\pi(\cdot \mid x) \| \pi_{\mathrm{ref}}(\cdot \mid x))$. Then $\pi^\star_\beta(y \mid x) \propto \pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}$, meaning for any y, y' with $r(x,y) = r(x,y')$ the relative odds remain unchanged. Consequently, $\rho(\pi^\star_\beta) = \rho(\pi_{\mathrm{ref}})$ and $\Lambda(\pi^\star_\beta) = \Lambda(\pi_{\mathrm{ref}})$ for every $\beta > 0$. This identical invariance holds at every step of any multiplicative update $\pi_{t+1}(\cdot \mid x) \propto \pi_t(\cdot \mid x) \exp(\eta f(r(x, \cdot)))$.*

This closed form relies on the Gibbs variational principle, standard in RL [Peters et al., 2010] and in the LLM setting [Korbak et al., 2022, Kruszewski et al., 2026, Dymetman, 2026]. We restate it because it is the baseline against which all departures are measured, and its consequence for reward hacking, that the composition of the accepted set stays frozen at the base model's, is the prediction under test. As $\beta \to 0$, $\pi^\star_\beta \to \pi_{\mathrm{ref}}(\cdot \mid \mathcal{A}(x))$: the whole regularisation path is the reference model conditioned on acceptance, which is why nothing inside the accepted set can move. We verify this invariance numerically to floating-point precision (Section 5).

Why the per-prompt statistic. We use a per-prompt statistic rather than a pooled one. The pooled quantity $\sum_x \pi(\mathcal{H}(x) \mid x) / \sum_x \pi(\mathcal{A}(x) \mid x)$ is a $\pi(\mathcal{A}(x) \mid x)$-weighted average of the invariant per-prompt ratios, and RLVR raises those weights toward 1 at differing rates. Its total excursion along the filtering path is exactly $-\operatorname{Cov}_x\big(\pi_{\mathrm{ref}}(\mathcal{A}(x) \mid x),\, \rho_x\big) \,/\, \mathbb{E}_x[\pi_{\mathrm{ref}}(\mathcal{A}(x) \mid x)]$ (Appendix A), which is *positive* whenever pass rate and hack fraction are negatively correlated across prompts. That is the typical case, as prompts a model can solve are prompts where it relies on the leak less. A practitioner tracking the pooled $P(\text{hack} \mid \text{rewarded})$ will therefore see it climb under pure filtering, with nothing exploited. In our environment it rises by +0.057 under exact mirror descent while ρ and Λ remain static. We report per-prompt statistics throughout.

3 Decoding the departure from the filter baseline

In practice, RLVR executes first-order updates on a shared-parameter model, often with a minimal or entirely absent KL term. To understand how models diverge from the theoretical baseline, we characterize the resulting drift exactly.

Lemma 1 (Drift decomposition)**.** *Under exact-gradient flow $\dot\theta = \nabla_\theta J(\theta)$,*

$$\frac{d}{dt} \log \pi_\theta(y \mid x) = \frac{1}{N} \sum_{x' \in \mathcal{X}} \operatorname{Cov}_{y' \sim \pi_\theta(\cdot \mid x')} \Big[r(x', y'),\; K_\theta\big((x,y),(x',y')\big) \Big], \tag{1}$$

where $K_\theta((x,y),(x',y')) = \langle \nabla_\theta \log \pi_\theta(y \mid x), \nabla_\theta \log \pi_\theta(y' \mid x') \rangle$ is the empirical tangent kernel of the log-policy.

Proof. By the score-function identity, $\nabla_\theta J = \frac{1}{N} \sum_{x'} \mathbb{E}_{y'}[r(x',y') \nabla_\theta \log \pi_\theta(y' \mid x')]$, and since $\mathbb{E}_{y'}[\nabla_\theta \log \pi_\theta(y' \mid x')] = 0$, the inner expectation operates as the (componentwise) covariance of the scalar r with the vector $\nabla_\theta \log \pi_\theta$. Taking the inner product with the fixed vector $\nabla_\theta \log \pi_\theta(y \mid x)$ and utilizing the linearity of covariance in its second argument directly yields (1). □

Equation (1) shows that the log-probability of a response on a single prompt is influenced by the dynamics on *every* prompt, in proportion to how strongly reward covaries with tangent similarity there. It is an instantaneous identity and we apply it as such. Our experiments use Adam [Kingma and Ba, 2015] rather than pure gradient flow. For an update $\dot\theta = P \nabla J$ with fixed, positive definite P, (1) holds with K_θ replaced by $K^P_\theta((x,y),(x',y')) = \langle \nabla_\theta \log \pi_\theta(y \mid x),\, P\, \nabla_\theta \log \pi_\theta(y' \mid x') \rangle$. Adam's first moment makes the update history-dependent, so the instantaneous identity does not apply verbatim. The support argument survives, which is all Corollary 1 requires: Adam's state is per-coordinate, and its momentum is a linear combination of gradients that are already block-supported, so a per-prompt parameterisation keeps a vanishing cross-prompt term under Adam as under flow.

(Our kernel is the finite-width empirical kernel, not the constant infinite-width limit of Jacot et al. [2018].)

Corollary 1 (Per-prompt parameters cannot see universality)**.** *If $\theta = (\theta_x)_{x\in\mathcal{X}}$ with $\pi_\theta(\cdot \mid x)$ operating as a function of θ_x only, then $\nabla_{\theta_{x'}} \log \pi_\theta(y \mid x) = 0$ for $x' \neq x$. Consequently, $K_\theta((x,y),(x',y')) = 0$ off the diagonal, and* (1) *collapses solely to the $x' = x$ term. Indeed, $\nabla_{\theta_x} J = \frac{1}{N}\sum_y \pi_\theta(y \mid x)(r(x,y) - V(x))\nabla_{\theta_x} \log \pi_\theta(y \mid x)$ depends on θ_x alone, rendering the trajectory of block x an autonomous ODE. A per-prompt policy generates no u-dependence of its own, and can depend on m only through what it inherited structurally at initialisation.*

By framing prompt optimization through this lens, the tabular arm serves as a strict control rather than merely a baseline. The argument survives Adam, whose state is per-coordinate, but would not survive a global gradient-norm clip, which we consciously omit. Importantly, this corollary does not imply that the per-prompt policy remains perfectly static on the filter baseline. For a softmax policy evaluating two equally rewarded responses, exact gradient flow yields $\frac{d}{dt}\log(\pi_1/\pi_2) = (\pi_1 - \pi_2)(1 - J)$. Thus, the more probable of the two natively grows, a recognized rich-get-richer behavior inherent to softmax policy gradients [Mei et al., 2020, Lochab et al., 2026]. However, that specific drift is entirely prompt-local and completely blind to u, establishing it as a valid control rather than a confounding variable.

While Equation (1) decomposes the drift, it does not explicitly isolate $u = m/N$. Specialising the identity for a leak h and isolating the h-dependent terms reveals that the effective acceptance fraction driving the drift is both kernel- and policy-weighted:

$$u_{\text{eff}}(x,h) \;=\; \Big(\sum_{x'\in U} w_{x'}\Big)\Big/\Big(\sum_{x'} w_{x'}\Big), \qquad w_{x'} = K_{\theta,xx'}\,\pi_\theta(h \mid x'), \tag{2}$$

where $K_{\theta,xx'}$ represents the kernel entry coupling h at x to h at x', and U denotes the accepting set. The raw fraction m/N is recovered perfectly only when the cross-prompt kernel entries and the leak probabilities remain homogeneous across prompts. To operationalize this, we establish a homogeneity assumption.

Definition 3 (Coherent kernel model)**.** We express the kernel as $K_\theta = K_\theta^{\text{coh}} + R$ with

$$K_\theta^{\text{coh}}((x,h),(x',y')) = \kappa_{xx'}\,\mathbf{1}[y' = h] \;+\; a(x,x') \;+\; b(x,x',y'), \qquad \kappa_{xx'} = \begin{cases}\kappa_s & x' = x\\ \kappa & x' \neq x\end{cases}$$

where a does not depend on y' and b does not depend on h. The a term contributes zero covariance in (1), and the b term is common to both the leak and the correct answer, canceling out in the within-prompt log-odds. Thus, only (κ, κ_s) and the residual R survive (Appendix A). We define a response as *coherent at scales* (κ, κ_s) when the induced error in (3) from R is small relative to κp_h. We do not strictly bound R; the proof in Appendix A substitutes K_θ^{coh} and drops it, rendering Proposition 2 a directional rather than absolute prediction. Crucially, coherence describes *how a response is computed*, not string identity: a leak emitted by prompt-independent machinery is highly coherent, whereas correct answers that happen to coincide coincidentally across prompts need not be.

Proposition 2 (Universality threshold)**.** *Let h be coherent at scales (κ, κ_s), $\kappa > 0$, accepted on $U \subseteq \mathcal{X}$ with $|U| = m$, $u = m/N$, and let g_x be the correct answer at x, coherent at scales $(\kappa_g, \kappa_{s,g})$. Write $V(x') = \pi_\theta(\mathcal{A}(x') \mid x')$ and $\bar{V} = J(\theta)$. Then for $x \in U$, assuming $\pi_\theta(h \mid \cdot) \equiv p_h$, $\pi_\theta(g_\cdot \mid \cdot) \equiv p_g$, and $V(\cdot) \equiv \bar{V}$:*

$$\begin{aligned}\frac{d}{dt}\Big[\log \pi_\theta(h \mid x) - \log \pi_\theta(g_x \mid x)\Big] &= \kappa\, p_h\,(u - \bar{V}) \;-\; \kappa_g\, p_g\,(u_g - \bar{V})\\ &\quad + \frac{1-\bar{V}}{N}\Big[p_h(\kappa_s - \kappa) - p_g(\kappa_{s,g} - \kappa_g)\Big],\end{aligned} \tag{3}$$

where u_g represents the universality of g_x. If the correct answer is largely incoherent ($\kappa_g \approx 0$), the leading-order sign cleanly follows $u - \bar{V}$. Consequently, the leak is amplified when its universality exceeds the policy's current verifier pass rate, and suppressed when it falls below it. The structural corrections are $O(1/N)$ and negligible when $N\kappa/\kappa_s \gg 1$.

Equation (3) reveals three significant structural insights. First, the verifier's false-positive rate is entirely absent: a system can accept far more distinct wrong responses per prompt, radically increasing its static audit failure rate, while leaving (3) completely unchanged. Second, parameter

sharing is functionally two-sided. The same mechanism that amplifies a highly universal leak forces an idiosyncratic leak *below* the base model's intrinsic rate, as it receives negative advantage updates on every prompt where it is not explicitly accepted, a penalty a per-prompt policy naturally avoids. Third, this threshold shifts dynamically during training: $\bar{V}$ is literally $J(\theta)$, and $dJ/dt = \|\nabla J\|^2 \geq 0$ under gradient flow. Therefore, leaks of intermediate universality might be amplified early in training but suppressed later, ultimately leaving only $u \approx 1$ leaks as survivors. This provides an internal, mechanistic explanation for the delay, plateau, and collapse taxonomy previously observed empirically [Egashira et al., 2026].

4 Experimental design

To isolate these mechanisms, we designed an environment satisfying three conditions: (i) the full response distribution can be computed exactly, eliminating sample noise; (ii) leak universality and false-positive rates are independently manipulable; and (iii) a shared-parameter policy and a per-prompt policy can be initialized from the *identical* distribution.

Task and response space. Prompts are the $N = 64$ pairs (a, b) with $a, b \in \{2, \ldots, 9\}$. The correct response is the decimal string of the product $a \times b$, zero-padded to two digits. The response space is all 100 two-digit strings, so $\pi_\theta(\cdot \mid x)$ can be enumerated exactly for every prompt at every step. Policies are autoregressive over the two digits.

Verifier. The deployed verifier accepts the correct string plus k designated wrong strings per prompt. None of them is the correct answer to any other prompt, so every response falls into exactly one category: correct, leak, or wrong. Prompts are partitioned into blocks of size m, and all prompts within a block share identical leak strings. Varying m manipulates universality $u = m/N$ while holding the static false-positive rate fixed at $k/(|\mathcal{Y}| - 1)$; varying k sweeps the false-positive rate while pinning universality.

Base model. Our architecture is a 2-layer, 64-dimensional transformer [Vaswani et al., 2017] (101,706 parameters), supervised fine-tuned on a target distribution that blends the correct answer, a per-prompt competence level λ_x spread over $[0, 0.9]$, and a minimal contamination rate $\varepsilon = 0.03$ of leak demonstrations. This intentional contamination aligns with findings by Khalifa et al. [2026], who observed that a mere trace of contamination is sufficient to seed hacking behavior. Competence is assigned via a fixed permutation independent of the block structure, preventing difficulty from correlating with universality. The base model achieves an accuracy of 0.441 and a baseline hack fraction of 0.128. While the SFT target remains identical across conditions, the realized base model, itself a shared-parameter network, naturally fits a universal leak map slightly better than an idiosyncratic one, resulting in a minor base hack fraction variation of 0.015. To account for this, we track and report all metrics as drifts from each run's specific base model.

Matched policies. We pair the transformer with a tabular policy containing free per-prompt parameters utilizing the identical autoregressive factorisation, completely parameterising the response simplex. This tabular policy is meticulously initialized to reproduce the base transformer's starting distribution to a tolerance of 10^{-4}. Both models then undergo identical training objectives, utilizing identical learning-rate grids wherever applicable (Table 1). While they differ fundamentally in parameter sharing, they also unavoidably differ in parameterisation; we specifically address and isolate this latter difference in Section 5.5.

Protocol. Unless stated otherwise, we run exact-expectation policy gradient with the exact mean baseline. This mirrors the expectation-limit of RLOO [Ahmadian et al., 2024] and Dr. GRPO [Liu et al., 2025], which debiases GRPO [Shao et al., 2024]. Group-relative variants [Yu et al., 2025] and PPO [Schulman et al., 2017] (inside its clipping range) share this structure. We run 400 steps with Adam, across three to ten seeds per cell (Appendix B, Table 1). We report the operating point a practitioner would select: the smallest learning rate in the grid whose final verifier reward falls within 10^{-3} of the maximum. This tie-break is conservative for our hypothesis, since smaller steps drift less, yet several learning rates saturate the verifier at reward 1 with vastly different true accuracies. Section 5.4 further validates the findings using sampled GRPO (with group-standard-deviation normalisation) and plain SGD.

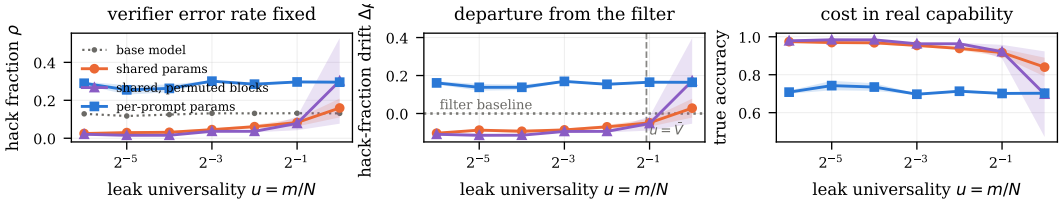


Figure 1: Universality, at a fixed verifier error rate, dictates whether RLVR suppresses a leak or amplifies it. Every verifier accepts exactly one wrong response per prompt, holding the static false-positive rate strictly identical while only varying the number of prompts sharing it. Left: Final hack fraction after training compared to the base model. Centre: Drift from the base model (theoretically zero under the filter baseline); the dashed line denotes $u = \bar{V}$. Right: Resulting true accuracy. The purple line repeats the sweep utilizing randomized block membership. Bands represent ± 1 s.e.m. across seeds.

5 Results

We report changes as drifts relative to each run's own base model: $\Delta\rho$ denotes the probability difference, $\Delta\Lambda$ the log-odds difference. Under the filter baseline of Proposition 1 both are exactly zero, making them immune to the residual base-model mismatch of Section 4. Absolute ρ is plotted alongside the base curve.

5.1 The filter baseline holds exactly, and gradient training departs from it

Our findings reveal that exact mirror descent preserves ρ invariance to 8.6×10^{-9} and Λ invariance to 9.6×10^{-8} over 400 steps. Similarly, the KL-regularised optimum preserves ρ invariance to 8.6×10^{-9} across $\beta \in [0.05, 2.0]$, holding to floating-point zero. Validating our analytical framework, Lemma 1 matches finite differences to a precision of 2.9×10^{-4}. We observe the per-prompt policy's cross-prompt kernel sits at exactly 0, starkly contrasting the transformer's 0.098 relative to its own self-kernel.

5.2 Universality dictates behavior for shared-parameter policies

Figure 1 is our primary finding. Holding the verifier's static false-positive rate constant and sweeping only the number of prompts that share the accepted wrong response, the shared-parameter policy's hack fraction climbs from 0.025 ± 0.005 at $u = 0.016$ to 0.160 ± 0.050 at $u = 1$ (10 seeds), against a base model value of 0.128. As a drift, $\Delta\rho$ moves from -0.104 ± 0.004 to $+0.028 \pm 0.050$, while true accuracy degrades from 0.974 to 0.840. The suppression at low u is substantial and unambiguous. The amplification at $u = 1$ under Adam falls within its own error bar, but the trend is a monotonic rise from strong suppression toward zero. A decisive sign change into active amplification appears under sampled GRPO and plain SGD (Section 5.4, Figures 4 and 6), using identical seeds, bases and step budgets and differing only in the gradient estimator or optimizer.

The outcome at $u = 1$ is heterogeneous across seeds rather than merely noisy. The ten final hack fractions span 0.02, 0.06, 0.06, 0.08, 0.08, 0.14, 0.17, 0.20, 0.22, 0.56, so the mean is skewed by a single collapsing seed; removing it leaves the drift at -0.017, still suppression. The heterogeneity is concentrated across prompts: at $u = 1$, a mean of 16% of prompts finish with the leak more probable than the correct answer, against 2% at $u = 0.016$. This is why $\Delta\rho$ approaches zero while $\Delta\Lambda$, dominated by the suppressed majority, climbs from -9.0 to -3.7 but stays negative. In the worst case a $u = 1$ run reaches hack fraction 1.000 at accuracy 0.000 while saturating the verifier.

Randomizing which prompts share a leak reproduces both the suppression and the slope (drift shifting from -0.110 to $+0.172 \pm 0.225$), so the effect is not an artifact of contiguous blocks coinciding with token structure, though its $u = 1$ endpoint here is carried by one of three seeds. That design also equalizes the base, constricting ρ variation to 0.0011 against 0.015 with contiguous blocks.

By contrast the per-prompt policy, initialized identically and trained under the same grid, remains flat. Regressing $\Delta\rho$ on $\log_2 m$ for the permuted sweep yields +0.0003 per doubling against +0.0353 for the transformer, two orders of magnitude apart; with contiguous blocks it yields +0.0028 against

+0.0177. The per-prompt policy does not sit *on* the filter baseline (moving +0.161 to +0.164), but its deviation carries no information about universality, which is what Corollary 1 requires. Section 5.5 eliminates the architecture confound.

In Appendix C, Figure 5 crosses universality against the static false-positive rate. Here, the base hack *mass* is strictly bounded to ε across all k. A two-way decomposition of the log-odds drift $\Delta\Lambda$ attributes a dominant 59% of its variance directly to universality ($u \in [0.125, 1]$), and only 11% to the false-positive rate across an eightfold range. Averaged over k, $\Delta\Lambda$ shifts drastically from -7.09 at $u = 0.125$ to -4.19 at $u = 1$; conversely, averaged over u, it barely shifts from -5.47 to -6.36 even as the false-positive rate multiplies eightfold. The directionality is critical: accepting *more* wrong responses per prompt does not inherently increase hacking and often mildly decreases it, as the base model's fixed propensity is diluted across rarer, individual strings. Standard audits that merely count total false positives are therefore measuring a metric that frequently exhibits both the wrong magnitude and the wrong sign.

5.3 Isolating the mechanism: cross-prompt gradient aggregation

Lemma 1 partitions the log-probability drift into an own-prompt term (present for all policies) and a cross-prompt term (the exclusive contribution of sharing). At initialization, the cross-prompt term sits at a negligible exactly 0 for the per-prompt policy across all universalities, proving Corollary 1 exact. For the shared policy, however, it climbs from 6.6 ± 1.1 at $u = 0.016$ to 12.2 ± 0.8 at $u = 1$, peaking sharply at 14.0 when $u = 0.5$. Coherence emerges organically from the architecture rather than being assumed: the mean pairwise cosine between per-prompt leak gradients grows from +0.02 to +0.29, while remaining at +0.03 for correct answers (though those answers also carry a universality of $u_g = 0.037$). Both trends are detailed in Figure 3 within Appendix C.

Limitations of observation. We were unable to pin down the exact $u > \bar{V}$ threshold pointwise dynamically. At initialization, the cross-prompt term remains positive globally due to a transient phase where all accepted response logits climb simultaneously. After 60 RL steps, this transient phase resolves, but $\bar{V}$ has simultaneously surged to 0.993. Consequently, the predicted threshold effectively hits 1, forcing the cross-prompt log-odds drift into small negative territory universally (-0.067 to -0.030). While this remains highly consistent with the proposition's behavior for $u \leq \bar{V}$, it precludes observing a clean sign change within the narrow observable window.

5.4 Robustness across GRPO, SGD, and KL penalties

Figure 4 in Appendix C covers GRPO and KL constraints, and Figure 6 (right) the SGD sweep. The vulnerability is not an artifact of exact-expectation gradients or of Adam. Under sampled GRPO (group size 8 with standard-deviation normalisation), $\Delta\rho$ shifts from -0.103 ± 0.004 to $+0.663 \pm 0.079$ across 10 seeds, with a definitive zero-crossing at $u = 0.27$. Plain SGD, the optimizer our analysis relies upon, produces the clearest separation in our study: the shared policy climbs from 0.051 to 1.000, with accuracy falling from 0.926 to 0.000, a slope of +0.184 per doubling and a crossing at $u = 0.07$. The per-prompt policy shifts only 0.053 to 0.055 (slope -0.001), staying *below* its base model throughout, which is the prompt-local rich-get-richer drift of Appendix A.

A KL penalty pulls the policy back toward the filter baseline, exactly as Proposition 1 suggests. Where resolved, however, it does so in a manner that *taxes* accuracy. Raising β to 0.3 forces $\Delta\rho$ from -0.103 back to -0.002 at $u = 0.016$, stripping away the *beneficial* suppression that parameter sharing provided, at a cost of 0.974 to 0.822 in true accuracy. KL regularisation is therefore not a free defensive patch; it commits the model to the base composition of the accepted set, which yields benefits only when that composition outperforms what targeted training would discover. This aligns with Ackermann et al. [2026], who conclude that biasing updates on reward accuracy outperforms blanket KL penalties.

5.5 Targeted controls: diagnosing the role of parameter sharing

The preceding sweeps contrast a deep transformer against a free per-prompt table, introducing a parameterisation confound alongside the sharing distinction. Figure 2 addresses this by introducing three targeted controls: one that entirely removes the parameterisation difference, and two that force the base model to remain structurally blind to the leak.

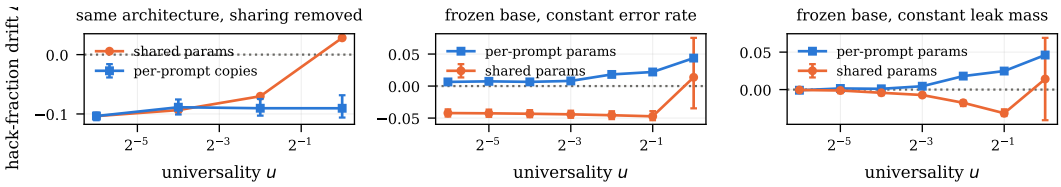


Figure 2: Left: substituting the shared model with independent per-prompt copies of the *identical* transformer architecture, under the same conditions and seeds. Centre & Right: freezing a structure-blind base model while varying only the acceptance mask, holding either the static false-positive rate or the accepted string's base mass constant. Bars denote bootstrap 95% intervals.

Architecture-fixed sharing. We assign each individual prompt its own complete, independent copy of the transformer architecture. All N copies are initialized identically from the exact same checkpoint and undergo identical optimization steps on the lowest three learning rates; crucially, copy x only ever evaluates prompt x. Under these conditions, the dependence on universality vanishes completely: the slope flattens to $+0.0019$ per doubling (CI $[-0.0008, +0.0050]$, consistent with absolute zero), contrasting sharply against $+0.0210$ for the shared model under matched conditions. Evaluating the paired difference within each individual seed yields $+0.0191$ (CI $[+0.0082, +0.0330]$, $t = 2.82$ on 10 seeds). At $u = 1$, the independent copies maintain a minimal drift of -0.090 and a robust accuracy of 0.959, whereas the shared model catastrophically fails on both metrics. This perfectly validates Corollary 1 while entirely stripping away the architecture confound.

Mask-only controls. Because altering the verifier alters the supervised target and therefore the base model, universality confounds with how deeply the base internalizes the leak (a structural coherence we measured growing seventeenfold). We therefore freeze one checkpoint per seed and vary only which prompts accept which of two fixed leak strings, in two variants that lock either the static false-positive rate or the accepted string's *per-prompt* base mass (Appendix B).

In neither blind variant does the shared policy show a steeper slope than the per-prompt baseline: marginal slopes of $+0.0056$ (CI $[+0.0003, +0.0121]$) and -0.0011 (CI $[-0.0074, +0.0051]$) against tabular baselines of $+0.0054$ and $+0.0073$, with paired differences of $+0.0001$ and -0.0084. Because this setup has the power to exclude an E10-sized effect, it is a definitive negative result rather than an underpowered null. These controls differ precisely where it matters: their base models remain *blind* to the leak structure. When the base encodes the pattern the slope manifests, and the independent copies control proves it requires sharing; when the base is blind, the slope mirrors the baseline.

This confirms that the vulnerability is intrinsically a two-stage mechanism: shared-parameter supervised training must first embed the leak's cross-prompt structure into the latent representation, after which shared-parameter RL actively amplifies the pattern it finds.

5.6 Strategic budgeting for verifier hardening

When assessing verifier security, two distinct patches that eliminate the exact same raw count of accepted wrong pairs, and so yield identical static audit improvements, generate wildly divergent impacts on model capability. Removing a single universal leak boosts the policy to 0.984 accuracy, whereas exhausting resources to remove every idiosyncratic leak only yields 0.583 accuracy. This represents a massive performance gap of 40 ± 16 points of true accuracy (3 seeds). The per-prompt policy inherently only displays a 11 point gap between these exact same verifiers (a gap already present in the base model), explicitly proving that the remaining 29 point delta is actively manufactured by RL. Experimental setup details are found in Figure 4 and Appendix C.

6 Related work

RLVR as filtering. The conceptualization of the KL-regularised optimum as an exponential-tilting form is well established [Korbak et al., 2022, Gai et al., 2025, Kruszewski et al., 2026, Dymetman, 2026, Lochab et al., 2026] and underpins the understanding that RLVR functionally sharpens, rather than broadly expands, base-model capability [Yue et al., 2025]. Notably, inverting

this exact relationship is how Rafailov et al. [2023] managed to completely bypass a separate reward model. While existing literature primarily interprets this invariance regarding the diversity among correct answers, we reposition it as a strict constraint on the internal composition of the accepted set, establishing it as our foundational null model. Similarly, we utilize the established rich-get-richer behavior among tied responses in softmax policy gradients [Mei et al., 2020, Lochab et al., 2026] as a baseline control.

Verifier error. Wang et al. [2020] analytically estimate the reward confusion matrix, proving theoretical convergence for policies trained on unbiased surrogate rewards, while Cai et al. [2026] derive explicit RLVR corrections that successfully outperform uncorrected training. Both critically demonstrate that *correction* aids performance, rather than suggesting uncorrected training is inherently robust. Separately, Plesner et al. [2026] directly measure noise tolerance, and Rad et al. [2026] formally map a phase transition based on the Youden index, showing incorrect modes face extinction when $J > 0$ but dominate severely when $J < 0$. Furthermore, spurious rewards have occasionally proven beneficial for specific model architectures [Shao et al., 2026]. Contemporary audits of verifier false positives [Zhao et al., 2026, Huang et al., 2026, Xu et al., 2025, Yuan et al., 2026] predominantly report gross error rates, largely ignoring the critical structural alignment of those accepted errors across varying prompts.

Systematic error. Egashira et al. [2026] previously established the foundational premise we expand upon: outcomes are dictated by the "specific pattern of introduced errors" rather than the blanket error rate, driven heavily by trigger frequency and conditional advantage at initialization. However, their triggers effectively operated at $u \approx 1$; in contrast, our work rigidly locks the error rate while systematically sweeping how many prompts share the accepted response. This identifies both the specific structural axis and the driving mathematical mechanism (a cross-prompt covariance that vanishes without sharing). It also reveals active *suppression* at low universalities, whereas prior outcomes were unilaterally non-positive. While Zhang [2026] points to inheritance, and Khalifa et al. [2026] or Nishimura-Gasparian et al. [2026] point to aggressive amplification and specification gaming, our framework harmonizes these findings. Unlike works defining what corrupted rewards *can* theoretically do [Everitt et al., 2017, Skalse et al., 2022, Gao et al., 2023], we decode exactly how standard optimizers mechanistically interact with them. Additionally, related studies on group-relative advantage estimators [Yang et al., 2026, Zeng et al., 2026] remain orthogonal, as our identified structural vulnerability persists even under exact-expectation gradients.

7 Conclusion

We established that while RLVR acts as a neutral filter at its KL-regularised optimum, cross-prompt parameter sharing breaks this baseline in practice. Building on the finding that the pattern of a verifier's errors dictates the extent of reward hacking [Egashira et al., 2026], we identified cross-prompt sharing as the structural mechanism driving it. Sharing lets a leak aggregate gradient updates from every prompt that accepts it while being penalized by every prompt that does not. Without shared parameterisation the amplification channel vanishes entirely: independent per-prompt copies of identical architectures exhibit zero dependency on leak breadth. Controls with structure-blind base models show the vulnerability is a two-stage process. Sharing at RL time is insufficient on its own; RL amplifies the cross-prompt structure already encoded during supervised training. The metric to audit is therefore not the gross frequency of verifier errors, nor the breadth of acceptance, but whether the admitted false positives align with the model's pre-existing shared error modes.

Limitations and Future Work. Proposition 2 assumes a structurally coherent kernel and constant pass rates; since our measured kernel variance fluctuated seventeenfold, the threshold remains a directional heuristic rather than an absolute bound. Exact enumeration required a two-token response space and designated leak strings, so string identity acts as a proxy for shared error modes, an assumption that must scale to open-ended real-world generations. We relied on a contaminated SFT base and three to ten seeds with heterogeneous $u = 1$ outcomes. Future work should determine whether these vulnerabilities translate to arbitrary open-ended tasks and whether these structural error modes remain invariant across frontier models. LLMs assisted with experiment design and manuscript preparation; every number comes from the released code.

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A Proofs

A.1 Proposition 1

Fix x. The objective $\sum_y \pi(y) r(x,y) - \beta \sum_y \pi(y) \log \frac{\pi(y)}{\pi_{\mathrm{ref}}(y)}$ over the simplex is strictly concave for $\beta > 0$; setting the derivative of the Lagrangian to zero gives $\pi^\star_\beta(y \mid x) = \pi_{\mathrm{ref}}(y \mid x) e^{r(x,y)/\beta} / Z_\beta(x)$ with $Z_\beta(x) = \sum_{y'} \pi_{\mathrm{ref}}(y' \mid x) e^{r(x,y')/\beta}$. For y, y' with $r(x,y) = r(x,y')$ the normaliser and the exponential factor cancel in the ratio. Since $r \geq c$ we have $\mathcal{H}(x) \subseteq \mathcal{A}(x)$ and $r \equiv 1$ on $\mathcal{A}(x)$, so

$$\frac{\pi^\star_\beta(\mathcal{H}(x) \mid x)}{\pi^\star_\beta(\mathcal{A}(x) \mid x)} = \frac{e^{1/\beta} \pi_{\mathrm{ref}}(\mathcal{H}(x) \mid x)/Z_\beta(x)}{e^{1/\beta} \pi_{\mathrm{ref}}(\mathcal{A}(x) \mid x)/Z_\beta(x)} = \frac{\pi_{\mathrm{ref}}(\mathcal{H}(x) \mid x)}{\pi_{\mathrm{ref}}(\mathcal{A}(x) \mid x)}$$

for every x, hence ρ and Λ (the latter whenever $\pi(\mathcal{H}(x) \mid x) > 0$) are unchanged. For a multiplicative update $\pi_{t+1}(y \mid x) \propto \pi_t(y \mid x) e^{\eta f(r(x,y))}$ the same cancellation applies at every step.

Excursion of the pooled statistic. Write $a_x = \pi_{\mathrm{ref}}(\mathcal{A}(x) \mid x)$, $\rho_x = \pi_{\mathrm{ref}}(\mathcal{H}(x) \mid x)/a_x$ and $s = e^{1/\beta}$. Then $\pi^\star_\beta(\mathcal{A}(x) \mid x) = s\, a_x / (1 - a_x + s\, a_x)$, and the pooled statistic is $P(s) = \sum_x \rho_x w_x(s) / \sum_x w_x(s)$ with $w_x(s) = s a_x / (1 - a_x + s a_x)$. At $s = 1$ (no RL) $P = \sum_x a_x \rho_x / \sum_x a_x$; as $s \to \infty$ every $w_x \to 1$ and $P \to \frac{1}{N} \sum_x \rho_x = \rho(\pi_{\mathrm{ref}})$. The total excursion is therefore

$$P(\infty) - P(1) \;=\; \frac{1}{N} \sum_x \rho_x \;-\; \frac{\sum_x a_x \rho_x}{\sum_x a_x} \;=\; -\frac{\mathrm{Cov}_x(a_x, \rho_x)}{\mathbb{E}_x[a_x]},$$

which is positive whenever pass rate and hack fraction are negatively correlated across prompts. The pooled statistic therefore drifts upward under pure filtering and is invariant only in the degenerate case $\rho_x \equiv$ const. It is not an estimator of ρ and we do not call it unbiased; it is a deterministic function of the filtering path that happens to converge to ρ in the strong-RL limit.

A.2 Lemma 1

$\nabla_\theta J = \frac{1}{N} \sum_{x'} \sum_{y'} \pi_\theta(y' \mid x') r(x', y') \nabla_\theta \log \pi_\theta(y' \mid x') = \frac{1}{N} \sum_{x'} \mathbb{E}_{y'}[r(x', y') \nabla_\theta \log \pi_\theta(y' \mid x')]$. Since $\mathbb{E}_{y'}[\nabla_\theta \log \pi_\theta(y' \mid x')] = 0$, the expectation equals $\mathrm{Cov}_{y'}(r, \nabla_\theta \log \pi_\theta)$, understood componentwise. For a fixed vector $v = \nabla_\theta \log \pi_\theta(y \mid x)$, $\langle v, \mathrm{Cov}(r, g) \rangle = \mathbb{E}[r \langle v, g \rangle] - \mathbb{E}[r] \mathbb{E}[\langle v, g \rangle] = \mathrm{Cov}(r, \langle v, g \rangle)$, so the inner product passes inside. □

A.3 Corollary 1

With $\theta = (\theta_x)$ and $\pi_\theta(\cdot \mid x)$ a function of θ_x only, $\nabla_{\theta_{x'}} \log \pi_\theta(y \mid x) = 0$ for $x' \neq x$, so every off-diagonal kernel entry is a sum of products with a zero factor. Moreover $\nabla_{\theta_x} J = \frac{1}{N} \sum_y \pi_\theta(y \mid x)(r(x,y) - V(x)) \nabla_{\theta_x} \log \pi_\theta(y \mid x)$ involves only θ_x, so the flow restricted to block x is autonomous and depends on the verifier only through $r(x, \cdot)$. □

Rich-get-richer under vanilla policy gradient. For a single prompt with free logits θ_y and $J = \sum_y \pi_y r_y$, $\partial J / \partial \theta_z = \pi_z (r_z - J)$, so for two responses with $r_1 = r_2 = 1$, $\frac{d}{dt} \log(\pi_1 / \pi_2) = \dot{\theta}_1 - \dot{\theta}_2 = (\pi_1 - \pi_2)(1 - J)$. The odds between equally rewarded responses are therefore *not* preserved by vanilla policy gradient, even with no parameter sharing; they move toward whichever response already has more mass, at a rate that vanishes only as $J \to 1$. This is the prompt-local drift referred to in Section 3; it is independent of u by Corollary 1.

A.4 Proposition 2

Two parts of the kernel drop out before Definition 3 is applied: any component of $K_\theta((x,h),(x',y'))$ that does not depend on y' contributes zero covariance in (1), and any component that does not depend on h is common to the leak and to the correct answer and cancels in the within-prompt log-odds. Substituting the definition, the covariance at x' is

$$\begin{aligned} \mathrm{Cov}_{y'}[r(x',y'), K_\theta((x,h),(x',y'))] &= \kappa \Big(\pi_\theta(h \mid x') r(x',h) - \pi_\theta(h \mid x') V(x') \Big) \\ &= \kappa \, \pi_\theta(h \mid x') \big(r(x',h) - V(x') \big), \end{aligned}$$

using $\mathbb{E}_{y'}[r] = V(x')$, with κ replaced by κ_s in the $x' = x$ term. Since $r(x',h) = \mathbf{1}[x' \in U]$, for $x \in U$,

$$\begin{aligned} \frac{d}{dt} \log \pi_\theta(h \mid x) = \frac{\kappa}{N} \Big[\sum_{x' \in U \setminus \{x\}} \pi_\theta(h \mid x')(1 - V(x')) - \sum_{x' \notin U} \pi_\theta(h \mid x') V(x') \Big] \\ + \frac{\kappa_s}{N} \pi_\theta(h \mid x)(1 - V(x)). \end{aligned}$$

With $\pi_\theta(h \mid \cdot) \equiv p_h$ and $V \equiv \bar{V}$ the bracket is $p_h[(m-1)(1-\bar{V}) - (N-m)\bar{V}] = p_h[N(u - \bar{V}) - (1 - \bar{V})]$, giving $\kappa p_h (u - \bar{V}) + \frac{p_h(1-\bar{V})}{N}(\kappa_s - \kappa)$. The identical computation for g_x with scales $(\kappa_g, \kappa_{s,g})$ and universality u_g gives $\kappa_g p_g (u_g - \bar{V}) + \frac{p_g(1-\bar{V})}{N}(\kappa_{s,g} - \kappa_g)$; subtracting yields (3). □

Validity of the leading-order threshold. The $O(1/N)$ terms are negligible when $N\kappa/\kappa_s \gg 1$. We measure κ/κ_s as the mean cross-prompt gradient cosine for the leak, and it is not constant: +0.29 at $u = 1$ gives $N\kappa/\kappa_s \approx 18$, but +0.02 at $u = 0.016$ gives $N\kappa/\kappa_s \approx 1.1$, at which point the correction is of the same order as the leading term. The leading-order threshold is therefore an argument about the high-universality end only; the suppression we observe at low u is an empirical finding, not a prediction of (3). Proposition 2 also treats κ as a constant while we measure it growing roughly seventeenfold with u, so the observed growth of the cross-prompt drift is jointly attributable to $\kappa(u)$ and to $(u - \bar{V})$ and the proposition does not separate them. Both are reasons we do not read a numerical boundary off the experiments.

Reading the threshold. $\bar{V}$ is the policy's *verifier* pass rate, observable during training without any ground truth. A leak is amplified when it is accepted more often than the policy currently passes. The statement is directional and first-order: it identifies the sign of the leading-order drift and the quantity controlling it, not an exact boundary in a model whose kernel is only approximately coherent.

B Environment, model and training details

Task. Prompts are all (a, b) with $a, b \in \{2, \ldots, 9\}$ ($N = 64$); the correct response is the two-digit decimal string of ab (all products lie in $[4, 81]$). The response space is all 100 two-digit strings. Reducing the operand set to $\{2, \ldots, 5\}$ gives $N = 16$ for the prompt-set-size sweep, which uses three seeds and the learning-rate grid $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ shared by both arms. The correct

answers are not perfectly prompt-specific (2×6 and 3×4 both give 12), so they carry a universality of their own, $u_g = 0.037$. Under Definition 3 what matters is measured coherence rather than string identity, and we find the correct responses to be near-incoherent across prompts (+0.03) while the leak is not. u_g is small and non-zero, and we report it rather than assume it away.

Leak assignment. Leak strings are drawn without replacement from the 69 two-digit strings that are not the correct answer of any prompt, so a leak is never a correct answer anywhere and every response is unambiguously classified. Prompts are partitioned into N/m blocks; each block receives k leak strings shared by all its prompts. The verifier's static false-positive rate is exactly $k/99$ in every condition, independent of m.

Blocks are contiguous in the default configuration, and with prompts enumerated first-operand-major a contiguous block of size $m \geq 8$ is exactly a set of first operands, a property the transformer can read off a single input token. That would confound universality with how simple the prompt-to-leak map is to represent, so we run the whole sweep a second time with block membership given by a random permutation of prompts and leak strings drawn independently per (m, k, seed) rather than nested across m. Both versions appear in Figures 1 and 3.

What the base model does and does not control. The SFT target puts identical hack mass on every prompt in every condition, but the realised base model is a shared-weight transformer that fits a universal leak map slightly better than an idiosyncratic one, so the measured base hack fraction still varies by 0.015 across universality conditions. We therefore report *drifts* from each run's own base model, $\Delta\rho$ and $\Delta\Lambda$, which Proposition 1 makes exactly zero under the filter baseline and which are immune to that residual mismatch. Absolute ρ is shown alongside the base curve so both are visible.

In the universality×error-rate factorial an additional correction is needed. The uniform component of the SFT target would otherwise place mass on each of the k leak strings, so base hack mass would grow with k and the two factors would not be separable. For that experiment the uniform component is spread only over responses that are neither the correct answer nor a leak, which makes total base hack mass exactly ε for every k. The measured base *fraction* still varies, mean $\rho = 0.128$ with spread 0.054 across runs, because diverting the uniform floor also changes the mass on the correct answer; it is the mass, not the fraction, that the design holds fixed. This is the one place where the SFT target differs between experiments, and it is the version the factorial requires.

Base model. A decoder-only transformer with 2 layers, model dimension 64, 4 heads and a 10-way output head over digits (101,706 parameters) is trained for 600 Adam steps at 3×10^{-3} on the exact target $q(y \mid x) = (1-\varepsilon)[\lambda_x \mathbf{1}[y = g_x] + (1-\lambda_x)\mathrm{Unif}(\mathcal{Y})] + \varepsilon\,\mathrm{Unif}(\mathcal{H}(x))$ with $\varepsilon = 0.03$. Competence levels λ_x are an evenly spaced grid on $[0, 0.9]$ assigned by a fixed permutation with a seed independent of the block structure, so difficulty and universality are uncorrelated by construction. Because the loss is against an exact target rather than samples, the resulting base distribution is matched across universality conditions without any per-condition calibration.

Tabular policy. Free parameters $w_1 \in \mathbb{R}^{N\times 10}$ and $w_2 \in \mathbb{R}^{N\times 10\times 10}$ with $\pi(y \mid x) = \mathrm{softmax}(w_1[x])_{y_1} \cdot \mathrm{softmax}(w_2[x, y_1])_{y_2}$. This factorisation is complete: 10 free first-token logits plus 10×10 free second-token logits parameterise the entire 100-dimensional simplex for each prompt. Given the base transformer's exact log-probabilities we set $w_1 = \log \sum_{y_2} \pi(y \mid x)$ and $w_2 = \log \pi(y \mid x) - w_1$, which reproduces the base distribution to within 10^{-4} in max-norm (asserted at run time).

RL. The default objective is the exact-expectation policy gradient with the exact mean baseline, $\nabla_\theta J = \frac{1}{N} \sum_x \sum_y \pi_\theta(y \mid x)\,[r(x, y) - V(x)]\,\nabla_\theta \log \pi_\theta(y \mid x)$, implemented as a weighted cross-entropy with detached weights. This is the infinite-group limit of RLOO and Dr. GRPO and removes sampling noise from the comparison. The sampled-GRPO experiment uses group size 8, group-standard-deviation normalisation and one inner epoch. Optimiser Adam (default) or SGD, 400 steps, learning-rate grid $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ for Adam and $\{0.3, 1, 3\}$ for SGD, applied identically to both policy classes in the sweeps where a grid is shared (Table 1; E3 and E5 use a fixed neural/tabular pair, E6 two). The reported operating point is, for each policy class and cell, the smallest learning rate whose final verifier reward is within 10^{-3} of the best, which is what a practitioner without ground truth would select (strict argmax in E6).

The mask-only controls (E9, E9b). One checkpoint per seed serves every universality condition. Its supervised target is contaminated with a shared string h and a per-prompt string s_x at every prompt, in identical proportion in all conditions, so the initial policy, the probability on each leak string and the initial tangent kernel are bit-identical across the sweep; we verify the base probability on h agrees to six decimals. The verifier then accepts h on the m prompts in U and, in E9, s_x on the rest, giving exactly N accepted wrong pairs and a static false-positive rate of exactly $1/(|\mathcal{Y}| - 1)$ in every condition. E9b accepts nothing wrong outside U, so every accepted wrong string is h and its base mass is identical across conditions, at the cost of a static false-positive rate equal to $u/99$ and therefore rising with u. Note also that only the *per-prompt* mass on the accepted string is held fixed in E9b: the aggregate accepted-wrong base mass, and with it base ρ, still scale with u because fewer prompts accept anything wrong at small m. Both that and the rising rate bias E9b's slope downward, which is why the shared-versus-per-prompt comparison is made within the same base rather than across conditions. The rising rate which if anything biases E9b's slope downward given that a higher rate mildly reduces hacking in our factorial. U is nested across m, so adjacent conditions differ only by which prompts switched.

The residual gradient in E9 is that the accepted string's base mass rises from 0.0098 to 0.0149: a shared-parameter SFT represents one string demonstrated at every prompt better than N distinct ones. We tried to remove it by reweighting the two families in the supervised target until their realised probabilities matched, iterating a damped fixed point. The mean gap closes to 6×10^{-4} but the per-prompt gap does not, staying near 6×10^{-3} against probabilities of about 1.2×10^{-2}: the shared network cannot represent an arbitrary per-prompt split. Universality and representational ease are therefore not independently manipulable in a shared-parameter model, since sharing is the common cause of both, which is why we bracket the confound with two designs rather than eliminating it in one. E9's and E9b's per-prompt arms responding to that residual gradient is what Corollary 1 permits, since the corollary rules out u-dependence generated by the dynamics, not u-dependence inherited at initialisation.

Per-prompt weight copies (E10). N independent copies of the same transformer, each initialised from the shared base checkpoint and trained on its own prompt alone with the same objective and optimiser, on the three smallest learning rates of the shared arm's grid. Copy x only ever evaluates prompt x, so the copies are independent and training them in one loop is equivalent to training them separately. The matched shared-parameter arm it is compared against is E1 restricted to the same four universality conditions and the same ten seeds, with slopes averaged per seed, which is why its slope (+0.0210) differs slightly from the seven-condition figure quoted in Section 5 (+0.0177).

Full specification. Prompts are serialised as the token sequence $[\text{BOS}, a, b, y_1]$ over a vocabulary of 12 symbols (10 digits plus BOS and a padding symbol), with learned token and learned absolute positional embeddings, a causal mask, pre-norm blocks, 4 heads, feed-forward width $4d = 256$ with GELU, final `LayerNorm`, and a 10-way output head shared across the two response positions. Embeddings are initialised $\mathcal{N}(0, 0.02^2)$ and the remaining parameters use PyTorch defaults. The correct response for prompt (a, b) is the decimal string of the *product* $a \times b$, zero-padded to two digits, so 2×3 has correct response `06`; all products lie in $[4, 81]$ and the response space is every two-digit string `00`–`99`. Optimisation is full-batch over all N prompts with no minibatching and no gradient clipping. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ and zero weight decay; SGD uses no momentum. A run's seed simultaneously sets the transformer initialisation, the leak-string draw and, in E4, the rollout sampling; the competence permutation is drawn from a separate fixed seed so difficulty is never correlated with the leak structure.

The default objective is exact-expectation policy gradient, $\nabla_\theta J = \frac{1}{N} \sum_x \sum_y \pi_\theta(y \mid x)[r(x, y) - V(x)]\nabla_\theta \log \pi_\theta(y \mid x)$, with $V(x) = \sum_y \pi_\theta(y \mid x) r(x, y)$ the exact mean baseline. E4 replaces it with sampled GRPO: $G = 8$ responses are drawn per prompt at temperature 1 from the full enumerated distribution, advantages are $(r_i - \text{mean})/(\text{std} + 10^{-4})$ within each group, groups whose reward standard deviation is zero contribute zero advantage and hence no gradient, and one inner epoch is taken per batch with no importance ratio, no clipping and no reference-KL term, so the estimator is group-normalised REINFORCE rather than a ratio-clipped PPO objective. E5 adds $-\beta\, \text{KL}(\pi_\theta(\cdot \mid x) \| \pi_{\text{ref}}(\cdot \mid x))$ computed exactly in closed form against the frozen base, not estimated from samples.

Measurement. ρ, Λ and true accuracy are computed from the exact enumerated distribution $\pi_\theta(\cdot \mid x)$ over all 100 responses at all N prompts, at every logged step; they are not Monte-Carlo estimates, and the only stochasticity is over seeds (transformer initialisation, leak-string draw, and rollout sampling in the GRPO condition). Two probes are exceptions: the gradient decomposition and the coherence measurement average over a fixed random subsample of 24 and 12 prompts respectively, for cost.

Per-prompt log-odds are clipped to ± 20 before averaging, so a prompt whose leak mass has collapsed cannot dominate Λ. In the headline Adam sweep this binds on a median of 1.6% of prompts (worst case 18.8%). It binds far harder where the policy collapses outright, up to 81% under SGD at $u = 1$, but neither the SGD nor the GRPO sweep contributes a reported Λ, and on the sweeps that do the ordering of every reported series is unchanged at clipping thresholds of 20, 40, 60 and 200. The final metric of every run is recomputed in float64 from stored per-prompt masses, so the 10^{-8} floor used during online logging does not enter any reported number.

The two null-model checks in Section 5 agree to the last bit by construction: 400 steps of mirror descent at $\eta = 0.05$ reach exactly $\pi_0 e^{20r}$, which is the $\beta = 0.05$ endpoint of the KL grid. They are one computation checked two ways, not two independent checks.

Per-experiment protocol. Seeds, learning-rate grids and the operating-point rule vary between experiments; Table 1 records what was actually run.

	sweep	seeds	learning rates	notes
E1	universality, contiguous blocks	10	5, shared by both arms	headline sweep
E2	universality, permuted blocks	3	4, shared by both arms	base spread 0.0011
E3	universality $\times$ error rate	3	1 pair (10^{-3} / 3×10^{-2})	clean SFT floor
E4	sampled GRPO	10	3, shared by both arms	$G = 8$, std normalisation
E5	KL sweep	3	1 pair (10^{-3} / 3×10^{-2})	6 values of β
E6	verifier hardening	3	2	strict argmax on verifier reward
E7	prompt-set size $N \in \{16, 64\}$	3	3, shared by both arms	tests u versus m
E8	plain SGD	10	3, shared by both arms	robustness
E9	mask-only, constant error rate	10	5, shared by both arms	one frozen base per seed
E9b	mask-only, constant leak mass	5	5, shared by both arms	one frozen base per seed
E10	per-prompt weight copies	10	3	same architecture as the shared arm

Table 1: What was run in each experiment. Where a grid is shared by both arms the operating point is the smallest learning rate within 10^{-3} of the best verifier reward, chosen independently for each arm.

C Additional results

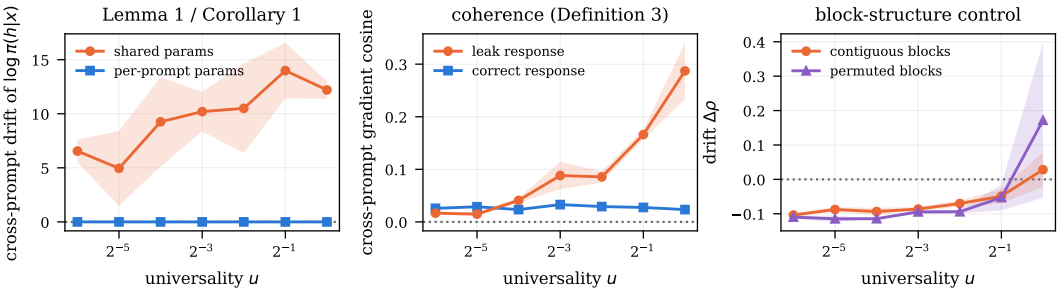


Figure 3: Mechanism. Left: the cross-prompt component of the exact drift of $\log \pi_\theta(h \mid x)$ at the base model (Lemma 1), identically zero for per-prompt parameters as Corollary 1 requires. Centre: mean pairwise cosine between per-prompt gradients, the coherence of Definition 3, measured rather than assumed. Right: contiguous versus randomised block membership.

Verifier hardening (E6). We give a verifier two leak families, one universal string plus one idiosyncratic string per prompt, and compare two patches that remove exactly the same number of accepted wrong (prompt, response) pairs and so improve any static audit identically. Removing the

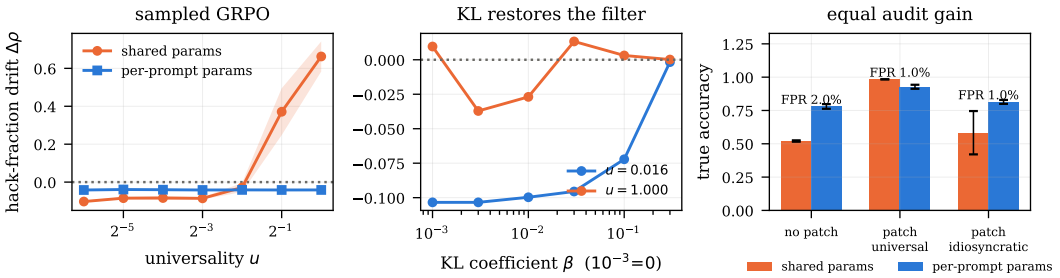


Figure 4: Practice. Left: sampled GRPO reproduces the pattern. Centre: a KL penalty pulls the policy back toward the filter baseline; the low-u side is resolved, the $u = 1$ side is within noise. Right: two verifier patches that improve the measured false-positive rate identically buy very different true accuracy.

universal leak leaves the policy at 0.984 true accuracy; removing every idiosyncratic leak instead leaves it at 0.583, a gap of 40 ± 16 points (3 seeds) from patches an audit rates as equally valuable (unpatched: 0.521). The per-prompt policy, which by Corollary 1 cannot respond to universality at all, shows a gap of only 11 points between the same two verifiers. That residue is what the base model already carried, since the SFT stage is itself shared-parameter (base hack fractions 0.064 and 0.094). The remaining 29 points are produced by RL.

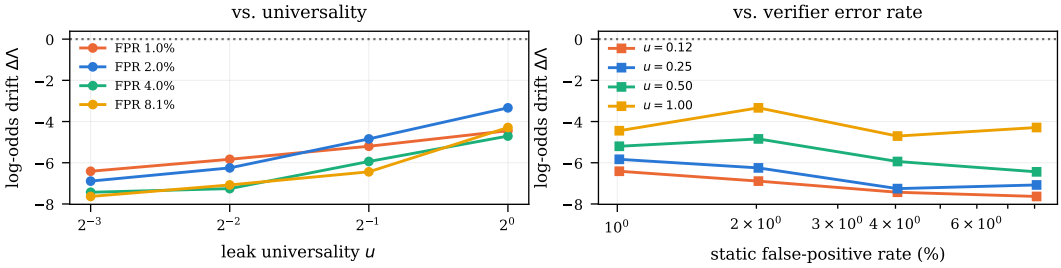


Figure 5: In this environment universality moves the drift and the verifier's error rate does not. Left: log-odds drift against universality, one line per false-positive rate. Right: the same data against the static false-positive rate, one line per universality, common y-scale. Varying accepted wrong responses per prompt from 1–8 moves the false-positive rate over 1.0–8.1% and mildly *reduces* hacking; universality moves the drift monotonically and much further.

Prompt-set size (E7). At fixed u the analysis predicts no dependence on the number of prompts. Repeating the sweep at $N = 16$ and $N = 64$ is consistent with that: at $u = 1$ the two agree (-0.004 ± 0.084 versus $+0.010 \pm 0.045$), whereas at the same block size $m = 16$ they sit at -0.079 against -0.004, a gap the three seeds do not resolve. Across the 5 matched values of u the largest gap is 0.065.

Figure 6 collects the four robustness views that do not fit in the main text: the universality effect across the whole Adam learning-rate grid rather than at the single reward-selected operating point, hack-fraction trajectories at the two extreme universality conditions for both policy classes, the sweep repeated at two prompt-set sizes, and the sweep under plain SGD.

D Reproducibility

All code, raw result files and figure scripts are included with the submission. A single machine with two CPU cores reproduces every experiment in the paper; no GPU is used. `src/env.py` defines the task and verifier, `src/policies.py` the two policy classes, `src/rlvr.py` the objectives and exact measurement, `src/theory.py` the numerical verification of Proposition 1 and Lemma 1, and `src/exp*.py` the experiments. Seeds 0–4 of E1 were produced by an earlier revision whose configuration records carry fewer fields. Re-running a cell from that half under the current revision

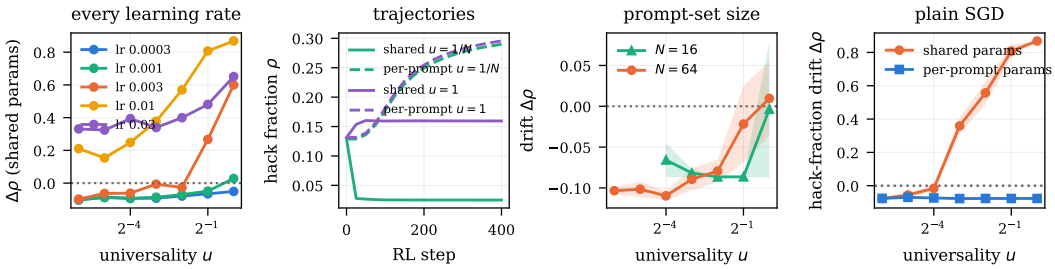


Figure 6: Left: the universality effect across the whole Adam learning-rate grid, rather than at the reward-selected operating point. Centre left: hack-fraction trajectories for the extreme universality conditions, both policy classes. Centre right: the sweep repeated at two prompt-set sizes, plotted against u. Right: the same universality sweep under plain SGD.

reproduces it to 10^{-8} and the stored base models are bit-identical, so the halves are pooled. One cell of E1 (the $m = 1$ condition) was run beyond the reported seed budget while a larger budget was being attempted; `src/figures.py` caps every E1, E4, E8, E9, E9b and E10 load at its reported seed budget and drops any seed not present at every universality condition, so the reported design is balanced at ten seeds for every universality condition rather than giving a few conditions more seeds than the rest. `src/figures.py` regenerates every figure and the macro file holding every measured number quoted in the text; the hand-entered constants are the step count, the contamination rate and the GRPO group size, which are settings rather than results, together with a small number of measured figures quoted in the text rather than through a macro: the clipping rates, $N\kappa/\kappa_s$, the seventeenfold growth of κ and its peak at $u = 0.5$, and the leave-one-seed-out drift of -0.017, all recomputed from the shipped result files, and the three probability gaps from the E9 reweighting attempt above, which come from a diagnostic run whose output is not shipped.